



Choosing the Vehicle: Taking an AI Equity Strategy On-Chain

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2026

Abstract

There are two live precedents for putting a fund on a blockchain and they are architecturally opposite. One is a registered closed-end interval fund whose token enforces transfer restrictions at the contract level. The other is a family of tokenized equities that are plain transferable tokens with compliance applied only at issuance and redemption. The first is offerable to United States retail investors and cannot touch decentralized finance. The second composes freely and is unavailable to United States persons.

That is not an accident of implementation. It is a constraint, and this paper states it as one: a token cannot simultaneously enforce holder-level restrictions and participate in permissionless protocols. Every design decision downstream follows from which side of that line a given instrument sits on.

We work through the consequence for a specific case — an artificial-intelligence equity strategy that already exists as a model portfolio — and conclude that the right answer is three separate instruments rather than one clever one: a fund for investors, a network token for the protocol, and a treasury for demonstrable assets. We give the registration path, the filing calendar and the operating cost of each, and we identify which of the three is unnecessary for most of what people want from the other two.

Contents

1	The strategy already exists	4
2	The constraint	4
3	The instrument menu	4
3.1	What the registered path actually carries	5
4	Three instruments, not one	5
5	Filing map	6
6	On-chain architecture	6
7	What we do not claim	7

1 The strategy already exists

The subject is concrete. An artificial-intelligence sleeve exists as model output over real market data: eight positions with model weights, carried alongside nine other asset-class sleeves in the same book.

Position	Name	Model weight	Tokenized equivalent
MSFT	Microsoft	19.1%	available
AVGO	Broadcom	14.3%	available
AMD	AMD	13.7%	available
SMH	Semiconductor ETF	13.7%	ETF class available
AMZN	Amazon	12.9%	available
NVDA	NVIDIA	10.5%	available
GOOGL	Alphabet	8.9%	available
META	Meta	6.7%	available

Table 1: The modelled sleeve. Weights are model output on real market data, not realized fund returns, holdings or net asset value.

The relevant fact is the right-hand column. A tokenized-equity programme that launched in 2025 with roughly sixty tickers has passed five hundred assets and \$35 billion of cumulative volume, covering United States stocks and exchange-traded funds one-to-one, settling on a public chain rather than through the conventional clearing system, trading continuously. Essentially every position in the sleeve has an on-chain representation today.

So the portfolio construction question is answered before we start. What remains is the wrapper, and the wrapper is where the entire difficulty lives.

2 The constraint

Definition 1 (Holder-restricted token). *A token whose transfer function consults an allowlist, a jurisdiction rule, or a holder registry, and reverts when the recipient fails it.*

Definition 2 (Permissionless composability). *The property that an arbitrary smart contract — an automated market maker, a lending pool, a vault — can hold and move a token without prior authorisation.*

Proposition 1 (No token has both). *Holder restriction and permissionless composability are mutually exclusive. A protocol contract is an address that no registry has vetted; a transfer to it either succeeds, in which case the restriction is not enforced, or reverts, in which case the token is not composable.*

This is not a limitation anyone has failed to engineer around. It is what enforcement means. The two live precedents sit on opposite sides of it and each is coherent on its own terms.

The practical corollary is worth stating bluntly because it is the mistake most plans make: *a token that is a registered fund share cannot be liquidity for anything.* Any design that requires both properties requires two tokens.

3 The instrument menu

Four doors, with what each costs and what each forbids. Figures are planning ranges for a fund of the size contemplated and require confirmation by counsel and by the service providers quoted.

	Registered fund token	Tokenized equity
Token standard	restriction-enforcing	plain transferable
Compliance point	every transfer	issuance and redemption only
Offerable to US retail	yes	no
Usable in a lending pool	no	yes
Bridgeable to another chain	only to permissioned venues	yes
Registration	investment company	none by the composer

Table 2: The two coherent designs. There is no third column.

Vehicle	Launch	Annual	Reaches	Composable
Registered interval fund	\$250–500k	\$700k–1.5M	US retail	no
Private fund, accredited	\$75–200k	\$150–400k	accredited US	restricted
Offshore fund, non-US	\$50–150k	\$100–250k	non-US	yes
Treasury, no fund	\$25–75k	\$50–150k	n/a	yes

Table 3: Vehicle cost and reach. Annual figures are recurring operating cost before any investment is made.

3.1 What the registered path actually carries

The heaviest door is worth itemising, because its cost is not a fee schedule but an institution. A registered closed-end fund operating as an interval fund requires a board with a majority of independent directors, a qualified custodian, a chief compliance officer and a written compliance programme, an annual audit by a registered public accounting firm, portfolio reporting monthly, an annual census filing, a separate filing for every periodic repurchase offer, a transfer agent, and post-effective amendments to keep the offering current.

It also imports a prohibition that is decisive here: a registered fund may not transact with its own affiliates as principal without an exemptive order. If the fund’s sponsor also issues a network token, the fund cannot buy that token and that token cannot be a claim on the fund. Any structure of the form *our token is backed by our fund* terminates at this sentence.

The live precedent for the registered path has been operating for six years on a portfolio of short-duration government securities and a money market fund, and is currently offering to repurchase six percent of its shares at net asset value. That is the structure functioning correctly. It is also a good measure of how much machinery is required to run the simplest possible portfolio inside it.

4 Three instruments, not one

The resolution is to stop trying to make one token do incompatible jobs.

The fund. An investable artificial-intelligence strategy, offered to investors, holding tokenized equities and liquid digital assets. Its token is restricted, its holders are known, and it is not composable. This is the instrument that takes outside capital.

The network token. Governance, staking and access for the protocol. Fixed supply, plainly transferable, bridgeable, poolable, usable as liquidity in another protocol. It is *not* a claim on the fund, and the difference between a claim and an association is the entire reason it can remain unrestricted.

The treasury. Assets held by the foundation, on-chain, at published addresses, verifiable by anyone. Not a pooled vehicle, not offered to investors, therefore not a fund and not registrable as one.

The treasury is the instrument most plans overlook, and for the specific goal of *demonstrable backing* it dominates the other two. A verifiable treasury is not a claim: holders can confirm the assets exist without holding a redeemable interest in them. That distinction is what keeps the network token unrestricted, and it is why the language used to describe it is load-bearing rather than cosmetic. Publishing an address and an amount is a statement of fact. Promising that a token is backed by that amount is an undertaking, and an undertaking of returns from a common enterprise is the classic securities test reciting itself.

	Fund	Network token	Treasury
Purpose	investable strategy	protocol utility	demonstrable assets
Holders	known, restricted	anyone	n/a
Composable	no	yes	yes
Redeemable	yes, periodic	no	no
Registration	by vehicle chosen	exemption or safe harbour	none
Annual cost	\$100k–1.5M	minimal	\$50–150k

Table 4: Division of labour. The relationships between the three are commercial, never collateral.

5 Filing map

What each path requires, in the order it is required.

Two sequencing notes. A registration statement removes the offering size limit entirely, so an exempt raise cap is irrelevant on that path and stacking the two for the same instrument is incoherent. And the exempt path for a non-security digital asset cannot carry an interest in a portfolio of securities, because its own predicate is that the asset is not itself a security — which a claim on a securities portfolio is.

6 On-chain architecture

For whichever vehicle ends up composable, the engineering is the same and is independent of the filing.

Vault. A tokenized vault holding three sleeves — tokenized equities tracking the model, liquid digital assets, and a rebalancing buffer — with in-kind creation and redemption for authorised participants and buffer-denominated subscription for everyone else.

Valuation across closed markets. The hard problem, and the one most implementations get wrong. Tokenized equities trade continuously; their underlying shares do not. Between Friday’s close and Monday’s open the token price is a forecast rather than a valuation. The vault’s oracle must carry an explicit stance for closed and halted markets rather than a stale last price, and any protocol using vault shares as collateral must widen its margin over exactly those windows.

Issuer concentration. A single issuer stands behind the entire tokenized-equity sleeve, holding the underlying shares through one legal entity. That is a concentration to diversify across issuers as alternatives mature, not to assume away.

Redemption asymmetry. The vault composes freely; converting its holdings back into registered shares still passes through the issuer’s compliance gate. Anyone modelling the position as fully liquid to cash is mistaken, and the disclosure should say so.

7 What we do not claim

Performance figures associated with this strategy are asset-class reference data or model output on real market data. They are not realized fund returns, net asset value or assets under management, and nothing here should be read as reporting any. The vehicles described are not offered by this document; any offering is made only by definitive documents to eligible investors. The cost ranges are planning estimates. The proposed crypto-asset regulation referred to above is a proposal open for comment and confers no exemption on anyone at the time of writing.

The structural conclusions — that holder restriction and composability cannot coexist in one token, that a registered fund cannot hold its affiliate’s token, and that a verifiable treasury is not a claim — are the load-bearing ones, and each should be confirmed by counsel before it is relied upon.

Path	Principal filing	Ongoing
Registered interval fund	registration statement for a closed-end fund	post-effective amendments; monthly portfolio reports; annual census; a filing per repurchase offer; annual audited financials
Private fund (accredited)	exempt-offering notice	state notice filings; adviser registration or exempt-reporting-adviser filing; annual audited financials to investors
Offshore fund (non-US)	none with the Commission	fund-domicile reporting; administrator and audit per offering documents
Network token	none today	if the proposed crypto-asset regulation is adopted: audited financials and periodic reports at the higher tier, and a certification to exit the investment-contract characterisation
Treasury	none	published addresses; periodic attestation

Table 5: Filing obligations by path. Form numbers deliberately omitted; the proposed crypto-asset regulation is a proposal and confers nothing until adopted.